

Host-to-Host communication MODELS

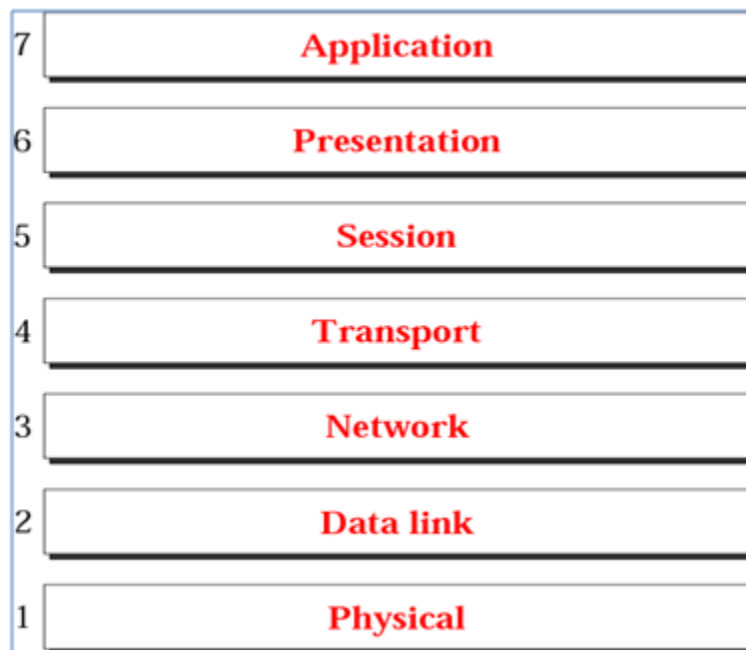
2.1 OSI Reference Model

What is an OSI model?

The OSI model or the OPEN SYSTEMS INTERCONNECTION (OSI) is a layered framework for the design of network system that allows for [communication](#) across all types of computer systems. It is designed by the INTERNATIONAL STANDARDS ORGANISATION (ISO) to define the basic standards for network communication. The OSI is a layered network model that explains step-by-step how systems communicate. A layered network model makes it easier to understand how packets are sent from one [computer](#) to another and reduces the complexity.

The OSI model consists of 7 [separate](#) but related layers, each of which defines a segment of process of moving information across a network.

7 [layers](#) of OSI model are:



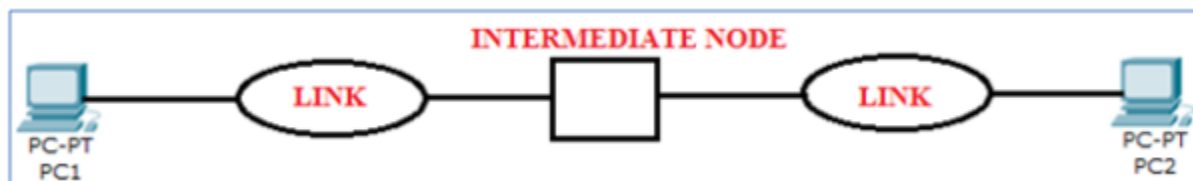
How to remember the names of the layers?

APPLICATION LAYER	- ALL
PRESENTATION LAYER	- PEOPLE
SESSION LAYER	- SHOULD
TRANSPORT LAYER	- TRY
NETWORK LAYER	- NEW
DATA LINK LAYER	- DIET
PHYSICAL LAYER	- PEPSI

The OSI is a layered architecture. What is meant by layered architecture?

The OSI model is built of 7 ordered layers.

Consider the figure below:



Various layers are involved when a message is sent from device A to device B. As the message travels from A to B, it may pass through many intermediate nodes. These intermediate layers usually involve only the first 3 layers of OSI model.

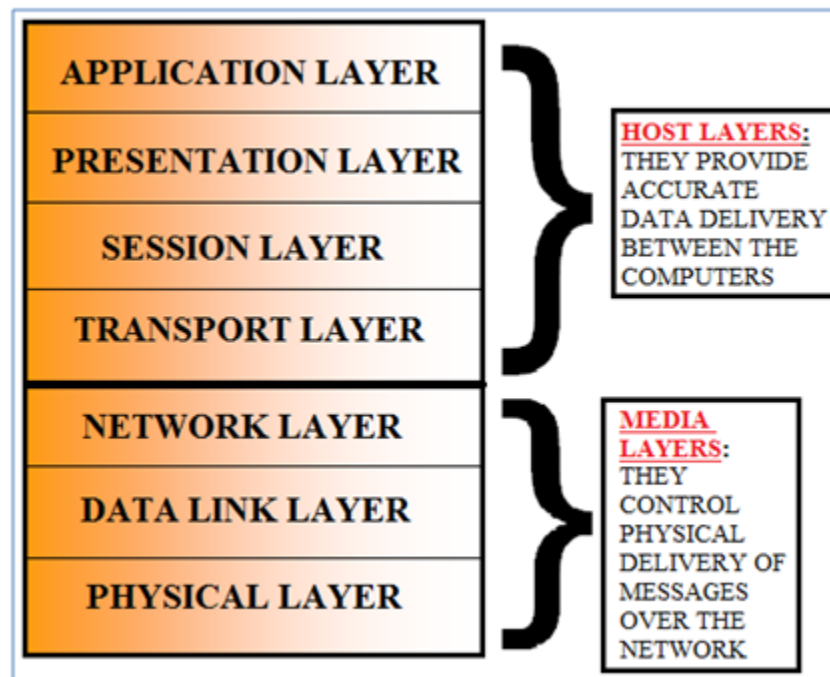
In developing the model, the designers distilled the process of transmitting data down to its most fundamental elements. They identified which networking functions had related uses and collected those functions into discreet groups that became layers. By defining and localizing functionally in this fashion, the designers created an architecture that is both comprehensive and flexible.

Advantages of layered architecture:

1. Allows you to modify a [component](#) without disturbing the next one.
2. Design scalable and maintainable application rapidly.
3. Increase security of an application.

The 7 layers of the OSI are divided into two [parts](#):

- The host layers: they provide accurate data delivery between the [computers](#).
- The [media](#) layers: they control physical delivery of messages over the network.



To understand how OSI model works, it is important to know the functions of each layer.

FUNCTIONS OF EACH LAYER:

7. APPLICATION LAYER:

This layer enables the user to access the network. It provides user interfaces and support for services such as e-mail, remote file access and transfer, shared database [management](#) and other types of distributed information services.

Functions of application layer are:

- It defines user interface tower level layers and various application processes.
- It [generates](#) and processes the requests that it receives.
- It deals with large amount of data, data transfer and user requirements for network.
- It is responsible for implementation of [security](#) checks at user entry points.

Specific services provided by the application layer are:

- **Network virtual terminal:** it is a software version of a physical terminal and allows a user to logon to a [remote](#) host.

- **File transfer, access and management (FTAM)**: this application allows a user to access files in a remote computer (to make changes or read data), to retrieve files from remote computer and to manage or control files in a remote computer.
- **Mail services**: this application provides basis for e-mail forwarding and storage.
- **Directory services**: this application provides distributed database sources and access for global information about various objects and services.

6. PRESENTATION LAYER: the presentation layer is responsible for inter-operability. On the sender side, it changes the information from its sender-dependent format to the common format and finally to the receiver dependent format on receiver side.

Various functions of this layer are

- It acts as an interpreter between the network and the node, it represents data according to the software/hardware environment of the node.
- It provides several functionalities related to data like:
 - **Data compression**: it reduces the number of bits to be transmitted by compressing the data.
 - **Data conversion**: it formats the data on different nodes as per their software/hardware environment.
 - **Data encryption**: it encodes data in a specific format so as it cannot be understood by every user or application.

5. SESSION LAYER:

It provides services related to sessions between two computers. Main services provided by this layer are:

- It allows users on different networks to establish a connection called 'session' between them.
- It provides services like authentication (username & password check), connecting (creating connection) & disconnecting (terminating connection) user access.
- It maintains the sequence of data packets and two-way communication.
- It provides a mechanism for a program to establish communication link with other program.
- It provides processes like synchronization so that data can be transmitted exactly from the point where the connection was broken.

4. TRANSPORT LAYER:

This layer is responsible for source-to-destination (end-to-end) delivery of entire message, whereas network layer oversees end-to-end delivery of individual packet without recognizing any relationship between those packets. The network layer treats each packet independently as though each piece belonged to a separate message, whether or not it does. The transport layer on the other hand ensures that the whole message arrive intact and in order, overseeing both error control and flow control at the source-to-destination level.

Responsibilities of transport layer are:

- **Service-point addressing**: computer often runs several programs at the same time. For this reason, source-to-destination delivery means delivery not only from one computer to the next but also from a specific process (running program) on one computer to a specific process (running program) on the other. The transport layer, therefore, must include a type of address called a service-point address (or, port address). The network layer gets each packet to the correct computer; the transport layer gets the entire message to the correct process on that computer.
- **Segmentation and reassembly**: a message is divided into transmittable segments, each segment containing a sequence number. These numbers enable the transport layer to reassemble the message correctly upon arriving at the destination and to identify and replace packets that were lost in transmission.
- **Connection control**: a transport layer can be either connection-oriented or connection-less. A connectionless transport layer treats each segment as an independent packet and delivers it to the transport layer at the destination machine. A connection-oriented transport layer makes a connection with the transport layer at the destination machine before delivering the packets. After all the data is transferred, the connection is terminated.
- **Flow control**: the transport layer is responsible for flow control. Flow control at this layer is performed end-to-end
- **Error control**: error control at this layer is performed end-to-end. The sending transport layer makes sure that the entire message arrives at the receiving transport layer without error (damage, loss or duplication). Error correction is usually achieved through re-transmission.
- It takes information from session layer and converts it into format understandable by other systems.

3. **NETWORK LAYER**:

It is responsible for the delivery of individual data packets from the source-to-destination across multiple networks. It adds the logical address of the sender and receiver to the packet. It also provides routing facility.

Functions of this layer are:

- **Logical addressing**: the network layer adds a header to the packet coming from the upper layers that among other things include the logical address of sender and receiver.
- **Routing**: when independent networks or links are connected together to create internetwork (a network of networks) or a large network, the connecting devices (called routers or gateways) route the packets to their final destination.
- It keeps the track of which layer receives the information.
- It translates the logical network address to physical machine address.
- It breaks larger packets into smaller ones so that they could be accepted by data link layer.
- It provides flow control and congestion control services.

2. DATA LINK LAYER:

It transforms the physical layer- a raw transmission facility to a reliable link and is responsible for node-to-node delivery. Its function is to move data frames from one station to other. Data link layer divides the stream of bits into number of frames. Also the physical address of the receiver device is added to the data frame. It provides error control mechanism to detect and retransmit corrupt or lost frame.

Functions of data link layer are:

- **Framing**: it defines the stream of bits received from the network layer into manageable data units called frames. It formalises data into equal units called frames and defines route for each frame.
- **Physical addressing**: if frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the physical address of the sender (source address) and/or receiver (destination address) of the frame. If the frame is intended for the system outside the sender's network, the receiver address is the address of the device that connects one network to the next. This means that data link layer gains physical access to network to transfer data.
- **Flow control**: if the rate at which the data is absorbed by the receiver is less than the rate at which data is transmitted by sender then data link layer imposes a flow control mechanism to prevent overwhelming the receiver.
- **Error control**: the data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames. It also uses the mechanism to prevent duplication of frames. Error control is normally achieved through a trailer added to the end of the frame.
- **Access control**: when 2 or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any given time.

1. PHYSICAL LAYER:

It co-ordinates the functions required to transmit a bit stream over a physical medium. It deals with the mechanical and electrical specifications of the interface and transmission medium. It also defines the procedures and functions that physical device and interfaces have to perform for transmission to occur.

Physical layer is concerned with the following:

- **Physical characteristics of interface and media**: the physical layer defines the characteristics of the interface between the devices and the transmission medium. It also defines the type of transmission medium.
- **Presentation of bits**: the physical layer data consists of a stream of bits (sequence of 0s and 1s) without any interpretation. To be transmitted, bits must be encoded into signals- electrical or optical. The physical layer defines the type of encoding (how 0s and 1s are changed to signals).

- **Data rate**: the transmission rate- the number of bits sent each second- is also defined by physical layer. This means that physical layer defines the duration of bits which is how long it lasts.
- **Synchronization of bits**: the sender and receiver must be synchronized at the bit level. In other words, the sender and receiver clocks must be synchronized.
- **Line configuration**: the physical layer is concerned with the connection of devices to the medium. Line configuration refers to the way two or more communication devices attach to a link. A link is a physical communication pathway that transfers data from one device to another. line configuration defines the attachment of communication devices to a link-
- **Point-to-point**: this configuration provides a dedicated link between two devices. The entire capacity of the channel is reserved for transmission between those two devices.
- **Multipoint**: a multipoint (also called multidrop) line configuration is one in which more than two specific devices share a single link.
- **Physical topology**: it defines how devices are connected to make a network. Devices can be connected using a mesh topology (every device connected to every other device), a ring topology (every device is connected to the next, forming a ring) or a bus topology (every device on a common link).
- **Transmission mode**: the physical layer also defines the direction of transmission between two devices: simplex, half duplex or full duplex.

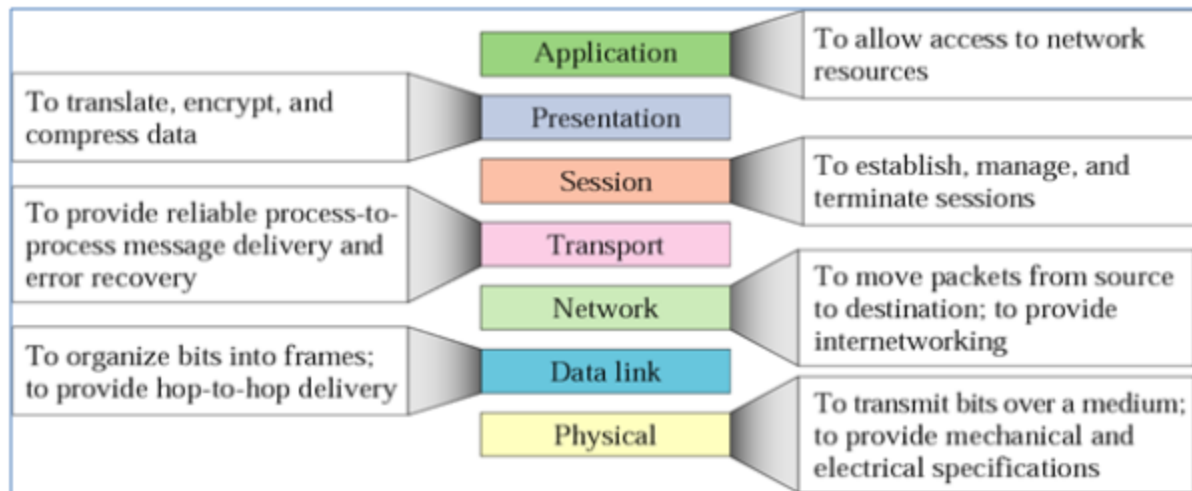
In the simplex mode, only one device can send, the other can only receive. The simplex mode is a one way communication.

In the half-duplex mode, two devices can send and receive but not at the same time.

In a full-duplex mode (or duplex mode), two devices can send and receive at the same time.

- The physical layer defines the electrical and mechanical specifications of cables and connectors.
- It also defines physical transmission medium between two stations.

SUMMARY OF LAYERS



HOW OSI MODEL HELPS IN COMMUNICATION?

The communication between two devices takes place step by step.

If a computer wants to send piece of information to another computer, following steps will take place:

1. The data will travel through APPLICATION LAYER which attaches its piece of information like provides network service.
2. Now data travels to presentation layer where data gets formatted i.e., encrypted, compressed or translated whatever is the requirement.
3. Now data travels down to the session layer where session layer performs its function of connection coordination.
4. Data then travels to transport layer where data is reassembled in sequence and any error is corrected.
5. Now data travels to network layer where data is routed to the destination using the IP address.
6. Data then travels to the data link layer where it is divided into frames. Data link layer uses the MAC address. This layer attaches additional information to the data. A **frame header** and **frame trailer** are attached to the data.

Frame header contains the source MAC address and the destination MAC address.

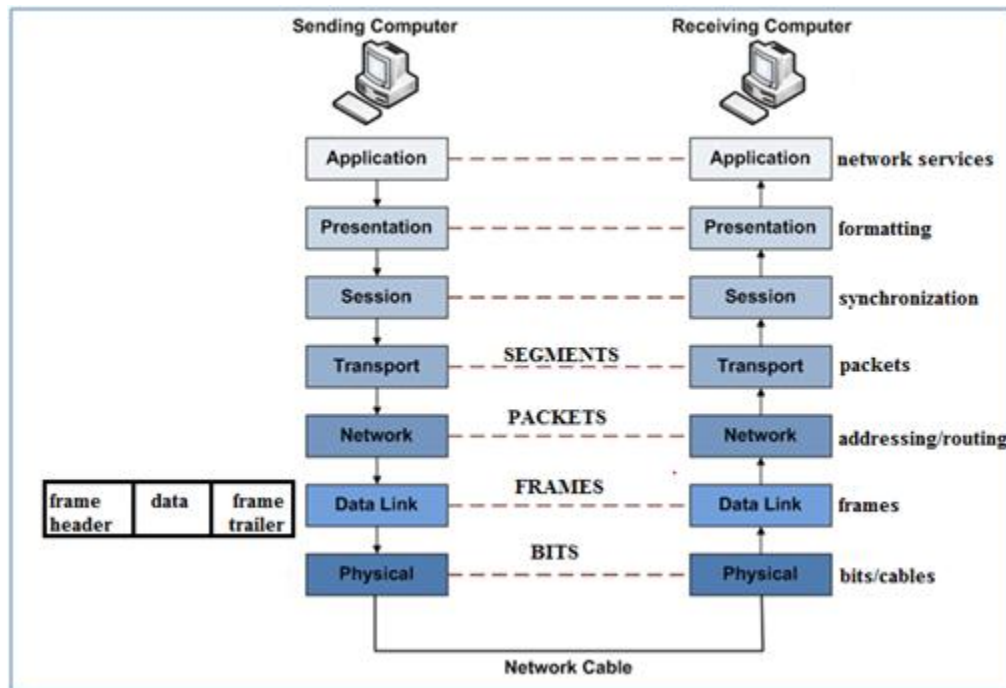
Frame trailer checks the frame loss or frame error.

7. At physical layer, data is transformed into bits and sent to the other device through the physical connection be it through cables or wireless.

When the receiving computer receives the data, following steps are followed:

1. The data is received by the physical layer now.

2. Then data travels to data link layer where the information is stripped out of the frame header and the frame trailer.
3. The network layer receives the data packets.
4. The transport layer receives data in sequence.
5. The session layer coordinates the communication.
6. The presentation layer decrypts the data which was encrypted by the presentation layer of source computer.
7. The application layer receives the data in readable form.



DIFFERENT DEVICES WORK ON DIFFERENT LAYERS:

1. **Gateways work on application layer.**
2. **Routers work on network layer.**
3. **Bridges/switches work on data link layer.**
4. **Hubs/ repeaters work on physical layer.**

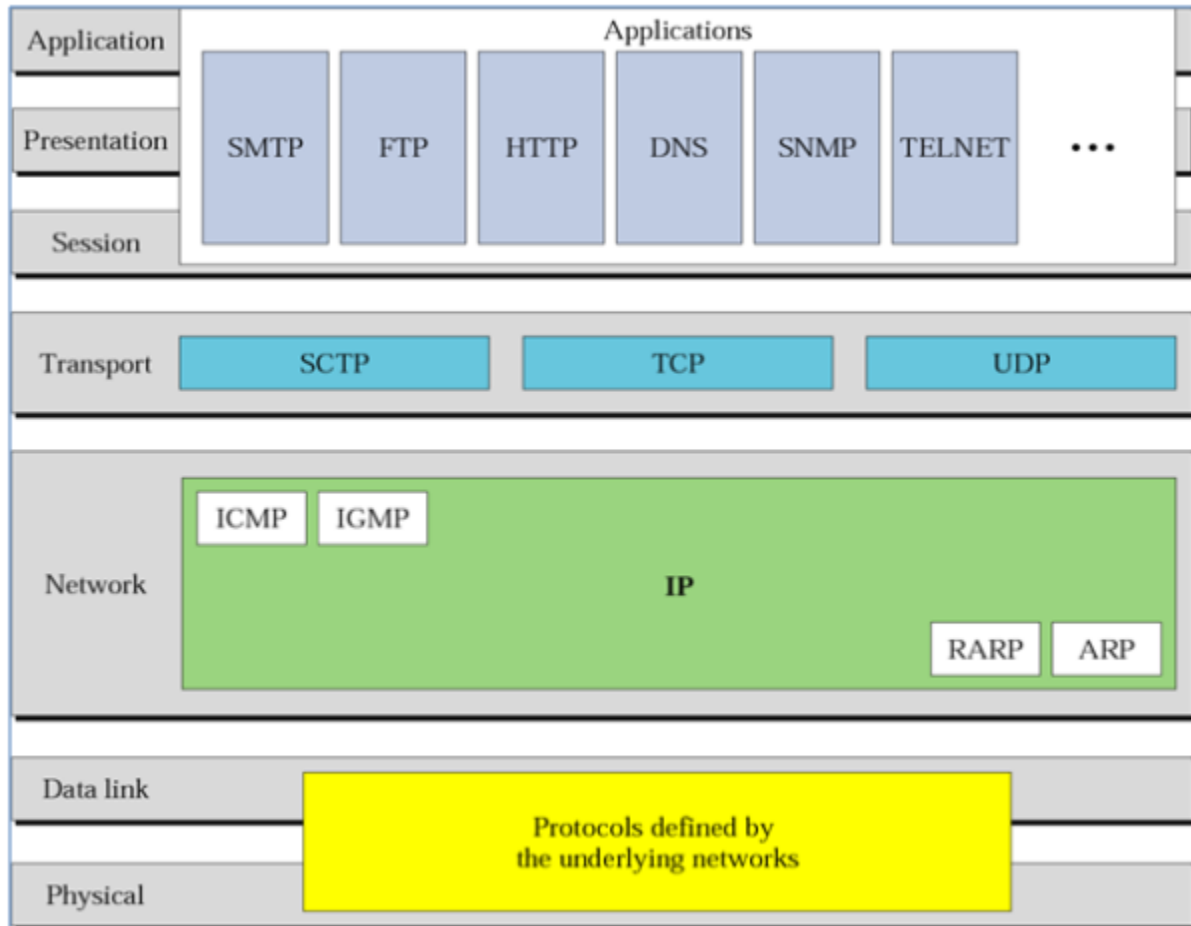
3.2 TCP/IP MODEL

The TCP/IP protocol suite is made of five layers:

- Physical layer
- Data link layer
- Network layer
- Transport layer

- Application layer

The first four layers provide physical standards, network interface, internetworking, and transport functions that correspond to the first four layers of the OSI model. The three topmost layers in the OSI model, however, are represented in TCP/IP by a single layer called the application layer.



Layer	Description	Protocols
Application	Defines TCP/IP application protocols and how host programs interface with transport layer services to use the network.	HTTP, Telnet, FTP, TFTP, SNMP, DNS, SMTP, X Windows, other application protocols
Transport	Provides communication session management between host computers. Defines the level of service and status of the connection used when transporting data.	TCP, UDP, RTP

Internet	Packages data into IP datagrams, which contain source and destination address information that is used to forward the datagrams between hosts and across networks. Performs routing of IP datagrams.	IP, ICMP, ARP, RARP
Network interface	Specifies details of how data is physically sent through the network, including how bits are electrically signaled by hardware devices that interface directly with a network medium, such as coaxial cable, optical fiber, or twisted-pair copper wire.	Ethernet, Token Ring, FDDI, X.25, Frame Relay, RS-232, v.35

TCP/IP Protocols

- **Hypertext Transfer Protocol (HTTP)**- Web servers implement this protocol. Short for Hypertext Transfer Protocol, the underlying protocol used by the World Wide Web. HTTP defines how messages are formatted and transmitted, and what actions Web servers and browsers should take in response to various commands. For example, when you enter a URL in your browser, this actually sends an HTTP command to the Web server directing it to fetch and transmit the requested Web page.
- **Simple Mail Transfer Protocol (SMTP)**- Used by e-mail servers (and sometimes Web servers) to send e-mail. Short for Simple Mail Transfer Protocol, a protocol for sending e-mail messages between servers. Most e-mail systems that send mail over the Internet use SMTP to send messages from one server to another; the messages can then be retrieved with an e-mail client using either POP or IMAP. In addition, SMTP is generally used to send messages from a mail client to a mail server. This is why you need to specify both the POP or IMAP server and the SMTP server when you configure your e-mail application.
- **Post Office Protocol Version 3 (POP3)**- A protocol used to retrieve e-mail from a mail server. Most e-mail applications (sometimes called an e-mail client) use the POP protocol, although some can use the newer IMAP (Internet Message Access Protocol). There are two versions of POP. The first, called POP2, became a standard and requires SMTP to send messages. The newer version, POP3, can be used with or without SMTP.
- **File Transfer Protocol (FTP)**- The protocol for exchanging files over the Internet. FTP works in the same way as HTTP for transferring Web pages from a server to a user's browser and SMTP for transferring electronic mail across the Internet in that, like these technologies, FTP uses the Internet's TCP/IP protocols to enable data transfer. FTP is most commonly used to download a file from a server using the Internet or to upload a file to a server (e.g., uploading a Web page file to a server).
- **DNS**- Short for Domain Name System (or Service or Server), an Internet service that translates domain names into IP addresses. Because domain names are alphabetic, they're easier to remember. The Internet however, is really based on IP addresses. Every time you use a domain name, therefore, a DNS service must translate the name into the corresponding IP address.

For example, the domain name <http://www.example.com> might translate to 198.105.232.4. The DNS system is, in fact, its own network. If one DNS server doesn't

know how to translate a particular domain name, it asks another one, and so on, until the correct IP address is returned.

- **Transmission Control Protocol (TCP)**- creates a reliable connection between two computers. TCP is one of the main protocols in TCP/IP networks. Whereas the IP protocol deals only with packets, TCP enables two hosts to establish a connection and exchange streams of data. TCP guarantees delivery of data and also guarantees that packets will be delivered in the same order in which they were sent.
- **User Datagram Protocol (UDP)**- Does not establish a connection, just sends messages. It is a connectionless protocol that, like TCP, runs on top of IP networks. Unlike TCP/IP, UDP/IP provides very few error recovery services, offering instead a direct way to send and receive datagrams over an IP network. It's used primarily for broadcasting messages over a network.
- **Internet Protocol (IP)**- Provides addressing scheme.. IP specifies the format of packets, also called datagrams, and the addressing scheme. Most networks combine IP with a higher-level protocol called Transmission Control Protocol (TCP), which establishes a virtual connection between a destination and a source. IP by itself is something like the postal system. It allows you to address a package and drop it in the system, but there's no direct link between you and the recipient. TCP/IP, on the other hand, establishes a connection between two hosts so that they can send messages back and forth for a period of time.
- **Internet Control Message Protocol (ICMP)**- Provides error messages. It is an extension to the Internet Protocol (IP). ICMP supports packets containing error, control, and informational messages. The PING command, for example, uses ICMP to test an Internet connection.
- **IMAP** – Short for Internet Message Access Protocol, a protocol for retrieving e-mail messages. The latest version, IMAP4, is similar to POP3 but supports some additional features. For example, with IMAP4, you can search through your e-mail messages for keywords while the messages are still on mail server. You can then choose which messages to download to your machine.

Encapsulation and De-encapsulation

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Peer to peer communication

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